



C#

Level 1

Explicit Casting



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Definition

Explicit Casting

-  Explicit casting is a manual conversion from one data type to another.

-  The target type is written in parentheses before the value: `(type)value`.

Explanation

Manual Conversion

-  `C#` does not automatically convert ``double`` to ``int`` because the decimal part may be lost.

-  ``(int)`` tells `C#` to convert the value to an ``int``.

Decimal Removal

-  Casting ``double`` to ``int`` discards the fractional part.

-  It does not round the number.

Procedures

1. Create a value with the original data type.
2. Specify the target type in parentheses.
3. Assign the converted value to the target variable.
4. Check for possible data loss.

💡 Example

```
double myDouble = 17.58;  
int myInt = (int)myDouble;  
Console.WriteLine("Double value: {0}", myDouble);  
Console.WriteLine("Int value cast from double: {0}", myInt);
```

➡ Result

Double value: 17,58

Int value cast from double: 17

Note

◆ ``17.58 → 17``

◆ ``17.99 → 17``

◆ ``17.50 → 17``

◆ Explicit casting removes the decimal part; it does not perform mathematical rounding.

◆ If actual rounding is required, use ``Math.Round()`` instead.

⊘ Warning

- ◆ `int myInt = myDouble;` is not allowed when `myDouble` is a `double`.
- ◆ Explicit casting can change or lose part of the original value.



Comparison

▨ Implicit Casting

- ✓ Automatic.
- ✓ No `(type)` required.
- ✓ Example: `double d = myInt;`

▤ Explicit Casting

- ✓ Manual.
- ✓ Uses `(type)value`.
- ✓ May cause data loss.
- ✓ Example: `int i = (int)myDouble;`

✦ Summary

- ✓ Explicit casting converts a value manually.
- ✓ Use `(targetType)value`.
- ✓ `double → int` discards the fractional part.
- ✓ Casting is different from rounding.

Thank you 😊